FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

# Birth Asphyxia & Hypoxic-Ischaemic Encephalopathy

Tutorial | Week 4 · Small babies, sugar & the heart | Neonatology | Dr Namugga

Birth asphyxia is failure to start or sustain breathing at birth because of interrupted oxygen delivery around labour; the brain injury that follows is hypoxic-ischaemic encephalopathy (HIE). Grade it with Sarnat staging, give meticulous supportive care (temperature, glucose, fluids, ventilation, seizure control with phenobarbital) and avoid secondary injury; in Uganda therapeutic cooling is not standard care outside tertiary NICUs.

**Must know before the ward round**

1. Birth asphyxia (WHO) - failure to initiate and sustain breathing at birth. HIE is the brain injury that follows; neonatal encephalopathy is the clinical syndrome (altered consciousness, tone, feeding, seizures).
2. Supporting criteria for an intrapartum hypoxic cause - cord/early arterial pH below 7.0 or base deficit 12 mmol/L or more, Apgar 0-3 beyond 5 min (or below 5 at 5 and 10 min), early encephalopathy, multi-organ dysfunction, a sentinel event.
3. Sarnat - mild (hyperalert, normal/increased tone, no seizures, usually normal outcome); moderate (lethargic, hypotonic, weak suck, seizures common); severe (stupor/coma, flaccid, absent reflexes, high death and disability).
4. Two waves of injury - primary energy failure at the time of the insult, then a latent phase of about 6 h, then secondary energy failure over 6-72 h. Treatment aims to prevent secondary injury.
5. Temperature 36.5-37.5 C - AVOID hyperthermia (worsens brain injury). Do not passively cool babies in Ugandan wards without a NICU protocol.
6. Therapeutic hypothermia (33.5 C for 72 h, started within 6 h, 36 weeks or more) reduces death/disability in high-income NICUs, but the HELIX trial (India, Sri Lanka, Bangladesh, 2021) showed no benefit and more deaths in LMICs.
7. Glucose - check early and often; treat below 2.6 mmol/L with 10% dextrose 2 mL/kg IV; avoid both hypo- and hyperglycaemia.
8. Fluids - start 10% dextrose 60 mL/kg/day on day 1 (often restricted to 40-60 mL/kg/day if oliguric); monitor urine output (oliguria below 1 mL/kg/h), weight and sodium (SIADH, acute kidney injury).
9. Seizures - phenobarbital 20 mg/kg IV over 15-20 min; extra 10 mg/kg doses to a total of 40 mg/kg; maintenance 5 mg/kg/day once daily. Check glucose and calcium; watch for apnoea.
10. Second-line anticonvulsants - phenytoin 20 mg/kg IV over 20-30 min, or levetiracetam 40-60 mg/kg IV, or midazolam; avoid repeated diazepam in neonates.
11. Prognosis - a baby still floppy or spastic, unresponsive and unable to suck at 1 week has severe brain injury and will do poorly (WHO); recovery of suck and normal tone by 1 week is reassuring.
12. Follow up every survivor of moderate/severe HIE for cerebral palsy, epilepsy, developmental delay, hearing and vision problems, and feeding difficulty.


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# In a nutshell

**Birth asphyxia** means the baby's oxygen supply was interrupted before, during or just after birth so that he fails to start or sustain breathing. The resulting brain damage is **hypoxic-ischaemic encephalopathy (HIE)**. "Neonatal encephalopathy" is the broader clinical term for a term baby with abnormal consciousness, tone, feeding or seizures; HIE is its commonest cause.

Why it matters in Uganda: intrapartum-related events (asphyxia) are one of the **three leading causes of neonatal death** (with prematurity and infection), a major cause of **fresh stillbirth**, and the commonest preventable cause of **cerebral palsy**. Most cases follow prolonged or obstructed labour, poorly monitored labour, cord prolapse, abruption, ruptured uterus or delayed resuscitation.

There is no antidote. In Ugandan units, the treatment is **meticulous supportive care**: keep the baby normothermic (and **never overheated**), keep glucose normal, give the right fluids, support breathing, **stop seizures with phenobarbital**, treat infection, feed carefully and follow up.

> **CLINICAL PEARL**
> The one thing never to forget: **prevention is the treatment.** A well-monitored labour (partograph, fetal heart every 30 min in active labour, timely action) and a trained person with a working bag and mask at every birth prevent most HIE.

# Definitions & classification

| Term | Meaning |
|---|---|
| **Birth asphyxia** (WHO) | Failure to initiate and sustain breathing at birth |
| **Perinatal asphyxia** (physiological) | Impaired gas exchange → **hypoxaemia, hypercapnia and metabolic acidosis** |
| **Neonatal encephalopathy (NE)** | Clinical syndrome in a baby of 35 weeks or more: reduced consciousness or seizures, abnormal tone and reflexes, poor feeding, often breathing difficulty |
| **HIE** | NE caused by intrapartum hypoxia-ischaemia |
| **Apgar score** | 0-10 at 1 and 5 min; describes the baby's state, **does not diagnose asphyxia** and is never used to decide whether to resuscitate |

**Criteria that support an intrapartum hypoxic-ischaemic cause** (ACOG/AAP "Neonatal encephalopathy and neurologic outcome"; used in research and medicolegal work):

1. **Cord (or early) arterial pH below 7.0** and/or **base deficit 12 mmol/L or more**
2. **Apgar 0-3 beyond 5 minutes** (or below 5 at 5 and 10 minutes)
3. Early-onset **moderate or severe encephalopathy**
4. **Multi-organ dysfunction** (kidney, liver, heart, gut, blood)
5. A **sentinel event** (cord prolapse, abruption, uterine rupture, shoulder dystocia) and fetal heart abnormalities
6. Neuroimaging consistent with acute hypoxic injury (basal ganglia/thalamus, watershed)

**UGANDA CONTEXT**

Cord blood gases are rarely available in Ugandan public hospitals. On the ward, asphyxia is diagnosed **clinically**: a history of a difficult or prolonged labour or fetal distress, a baby who needed resuscitation (low Apgar), and **encephalopathy** in the first hours to days. Write all three in your notes.

### Sarnat staging (modified): the ward version

| Feature | Stage 1 (mild) | Stage 2 (moderate) | Stage 3 (severe) |
|---|---|---|---|
| **Level of consciousness** | **Hyperalert**, irritable, jittery | **Lethargic** or obtunded | **Stupor or coma** |
| **Tone** | Normal or increased | **Mild hypotonia** | **Flaccid** |
| **Posture** | Mild distal flexion | Strong distal flexion | Intermittent decerebration |
| **Stretch reflexes** | Brisk | Brisk | Decreased or absent |
| **Moro** | Strong, low threshold | Weak, incomplete | Absent |
| **Suck** | Weak | Weak or absent | Absent |
| **Autonomic** | Sympathetic: **dilated pupils**, tachycardia | Parasympathetic: **constricted pupils**, bradycardia, secretions | Both depressed: pupils variable/poor light reflex, apnoea |
| **Seizures** | **None** | **Common** (often subtle) | Uncommon clinically (EEG suppressed) or frequent early |
| **Duration** | Under 24 h | 2-14 days | Hours to weeks |
| **Outcome** | **Usually normal** | **Variable**: about 20-40% die or have disability (lower with cooling in HICs) | **Death in about half or more; nearly all survivors disabled** |

**EXAM FAVOURITE**

Examiners love: "Mild = hyperalert + no seizures + normal outcome; moderate = lethargy + hypotonia + seizures; severe = coma + flaccid." Also: **pupils dilated in mild (sympathetic), constricted in moderate (parasympathetic)**.

The **Thompson score** (0-22; tone, consciousness, fits, posture, Moro, grasp, suck, respiration, fontanelle) is a simple bedside alternative, used in African and Asian research and some Ugandan units. A higher score means worse encephalopathy; scores of about 15 or more on days 1-7 predict poor outcome.

## Epidemiology & Uganda context

- Worldwide, intrapartum-related events cause roughly a quarter of neonatal deaths and over a million stillbirths a year, almost all in LMICs.
- In Uganda, many asphyxiated babies come from **prolonged or obstructed labour**, delayed referral from lower-level facilities, home births, inadequate fetal monitoring and a lack of skilled resuscitation.
- **Who is at risk?** Prolonged second stage, obstructed labour, malpresentation, cord prolapse, APH (abruption), ruptured uterus, eclampsia, maternal shock, breech and twin delivery, shoulder dystocia, thick meconium, oxytocin overuse (hyperstimulation), preterm and post-term babies.
- **Facilities**: HC III/IV should resuscitate (HBB) and refer; regional and national referral hospitals (Mulago/Kawempe, Mengo, Nsambya, Lubaga) have newborn units with oxygen, phenobarbital and IV fluids.

**Therapeutic hypothermia** is not standard in public Ugandan units; aEEG/EEG and MRI are available in only a few Kampala centres.

# Pathophysiology

## The two phases of brain injury

```
Insult (hypoxia + ischaemia)
   |
   v
PRIMARY energy failure  -> ATP falls, Na/K pumps fail, cytotoxic oedema,
   |                          glutamate release, calcium influx, cell necrosis
   v
Resuscitation -> partial recovery of brain energy
   |
   v
LATENT phase (~ 1-6 hours)  <- the "therapeutic window" (cooling must start here)
   |
   v
SECONDARY energy failure (6-72 h) -> mitochondrial failure, free radicals,
                                     inflammation, apoptosis, seizures
   |
   v
TERTIARY phase (weeks-months) -> ongoing injury, impaired repair
```

Everything that adds to the energy deficit in the secondary phase makes the injury worse: **hyperthermia, hypoglycaemia, hyperglycaemia, seizures, hypotension, hypoxaemia, hyperoxia, hypocapnia (over-ventilation reduces cerebral blood flow), infection and fluid overload**.

## The "diving reflex" and multi-organ injury

When oxygen falls, the fetus redistributes blood to the **brain, heart and adrenals** at the expense of the **kidneys, gut, liver, skin and lungs**. So asphyxia damages several organs:

| Organ | Consequence | What to monitor |
|---|---|---|
| **Brain** | HIE, seizures, cerebral oedema | Neuro exam, seizures |
| **Kidneys** | Acute tubular necrosis → **oliguria, AKI**, hyponatraemia | Urine output, creatinine, sodium |
| **Heart** | Myocardial dysfunction, tricuspid regurgitation, hypotension | HR, CRT, BP |
| **Lungs** | Meconium aspiration, persistent pulmonary hypertension (PPHN), haemorrhage | SpO2 (pre/post-ductal) |
| **Liver** | Raised transaminases, coagulopathy, hypoglycaemia | Glucose, bleeding |
| **Gut** | Feed intolerance, NEC | Abdomen, aspirates |
| **Blood** | Thrombocytopenia, **DIC** | Bleeding, platelets |
| **Endocrine/metabolic** | **Hypoglycaemia**, hypocalcaemia, SIADH | Glucose, calcium, sodium |

## Patterns of brain injury

- **Acute profound ("near total")** insult (e.g. cord prolapse, uterine rupture): deep grey matter (**basal ganglia, thalami**) → dyskinetic/spastic quadriplegic CP.

- **Prolonged partial** insult (e.g. long obstructed labour): **watershed** cortex and white matter → cognitive problems, spastic CP, cortical visual impairment.

# Clinical acumen

## Focused history

| Question | Why it matters |
|---|---|
| **Labour**: duration, augmentation, partograph, obstructed, malpresentation | Mechanism and severity of the insult |
| **Fetal heart**: bradycardia, decelerations, meconium | Evidence of fetal hypoxia |
| **Sentinel event**: cord prolapse, abruption, rupture, eclamptic fit, shoulder dystocia | Supports intrapartum cause, predicts pattern |
| **Resuscitation**: time to first breath/cry, bag-mask duration, HR response, compressions/adrenaline, Apgar 1/5/10 | Longer resuscitation = more severe HIE |
| **Gestation and birth weight** | Cooling eligibility (36 weeks or more); preterm encephalopathy has different causes |
| **Maternal fever, ROM duration, foul liquor** | **Sepsis/meningitis** can mimic or add to HIE |
| **Maternal drugs** (opioids, MgSO4, diazepam) | Can cause depression without HIE |
| **Seizures noticed**: when, what they looked like | Timing (HIE seizures typically start within 12-24 h) |
| **Feeding, urine output since birth** | Suck = neuro status; oliguria = AKI |
| **Maternal HIV (last VL), syphilis, blood group** | Prophylaxis and other causes |

## Examination

- **Vitals**: temperature (avoid above 37.5), HR, RR/apnoea, SpO2, CRT, BP if available.
- **Neurological exam (Sarnat)**: level of consciousness, spontaneous activity, tone (pull-to-sit, ventral suspension), posture, Moro, suck, grasp, pupils, breathing pattern, fontanelle.
- **Seizures**: subtle (lip smacking, eye deviation, cycling, apnoea), clonic, tonic, myoclonic.
- **Other organs**: chest (meconium aspiration, PPHN), heart (murmur of TR, poor perfusion), liver size, bleeding, abdomen, urine output.
- **Look for other causes**: dysmorphism, skin lesions, bulging fontanelle (meningitis), cephalhaematoma or subgaleal swelling (blood loss), jaundice.
- **Repeat the Sarnat exam** at least daily (and at 6 h if cooling were possible): encephalopathy evolves.

## Red flags

RED FLAG

- **Apnoea** or irregular breathing, SpO2 falling
- **Seizures**, especially repeated or prolonged; seizure in the first hours
- **Coma, flaccidity, absent suck and Moro** (severe HIE)
- **Hypotension**, poor perfusion, bradycardia
- **No urine for 24 h**, rising weight, oedema (AKI, fluid overload)
- **Bleeding** from puncture sites, gut or cord (DIC)
- **Hypoglycaemia** (below 2.6 mmol/L) or hyperthermia (above 37.5 C)
- **Bulging fontanelle** or fever (think meningitis)

### Thinking like a clinician

- **Not every floppy baby after a hard labour has HIE.** Always consider **hypoglycaemia, sepsis/meningitis**, maternal drugs, intracranial haemorrhage (subgaleal, subdural after instrumental delivery), metabolic disease, congenital neuromuscular disorders.
- **Encephalopathy without a hypoxic history** (normal labour, good Apgar) points away from HIE: look for infection, metabolic disease, stroke, or malformation.
- **Seizures in HIE are often subtle**: a baby with apnoea and eye deviation is seizing until proven otherwise.
- **Over-treatment harms too**: hyperthermia from radiant heaters, hyperoxia from 100% oxygen, fluid overload from "routine" 60-90 mL/kg/day in an oliguric baby, and stacked anticonvulsants causing apnoea.
- **Mild HIE today can be moderate tomorrow.** Re-examine; write the stage and the time.
- **Prognosis needs time.** Avoid firm predictions on day 1; recovery of suck and tone over the first week is the best clinical guide in Uganda.

## Clinical workup

### Step 0: First 5 minutes

1. **Resuscitation** as per HBB/neonatal resuscitation (see that topic): dry, airway, **bag-mask ventilation by 60 s**, HR checks, compressions/adrenaline in hospital.
2. Once breathing, **admit** if: prolonged ventilation (more than 1-2 min), abnormal tone, poor suck, seizures, or ongoing respiratory distress.
3. **Temperature**: aim **36.5-37.5 C**. Turn the radiant warmer down if above 37.5; do not overwrap. **Do not actively cool** without a cooling protocol and NICU.
4. **Airway and breathing**: position, suction if secretions; oxygen only to keep SpO2 **90-95%** (avoid hyperoxia); bag-mask for apnoea.
5. **Glucose** within the first hour and every 3-6 h for 24-48 h; treat below **2.6 mmol/L**.
6. **IV access** (peripheral or umbilical venous catheter); start **10% dextrose** at **60 mL/kg/day**.
7. **Seizure? → phenobarbital 20 mg/kg IV**.
8. **Sepsis risk** → blood culture, **ampicillin + gentamicin**.
9. **Vitamin K** if not given.
10. Call the senior; inform and support the mother.

### Step 1: Focused history

- [ ] Labour length, partograph, augmentation, obstruction: severity.
- [ ] FHR abnormalities, meconium, sentinel event: timing and pattern.
- [ ] Resuscitation details and Apgar 1/5/10 min: severity.
- [ ] Gestation and birth weight: other diagnoses; cooling eligibility where available.
- [ ] Infection risks: maternal fever, ROM over 18 h, foul liquor.
- [ ] Maternal medicines: opioids, MgSO4.
- [ ] Seizure description and timing.
- [ ] Urine and meconium passed.
- [ ] Maternal HIV/VL, syphilis, blood group; immunisation (BCG/OPV0).

### Step 2: Focused examination

- General: consciousness, activity, colour, posture, seizures.
- Vitals and perfusion (CRT below 3 s), urine output.

- **Sarnat stage** with each item recorded (consciousness, tone, posture, reflexes, Moro, suck, pupils, autonomic, seizures).
- Head: fontanelle, sutures, scalp swelling (caput vs cephalhaematoma vs **subgaleal haemorrhage**).
- Chest, heart, abdomen, liver; bleeding; skin.
- Document key negatives: "no bulging fontanelle, no dysmorphism, no subgaleal swelling".

### Step 3: Bedside tests

| Test | What you are looking for |
|---|---|
| **Glucose** | Below 2.6 mmol/L (treat); above 10 mmol/L (reduce glucose infusion rate) |
| **Temperature** | Keep 36.5-37.5 C; avoid above 37.5 |
| **SpO2** | 90-95%; check pre- and post-ductal if PPHN suspected (difference above 3-5%) |
| **Urine output** (weigh nappies: 1 g = 1 mL) | Below **1 mL/kg/h** after the first 24 h = oliguria |
| **Daily weight** | Weight gain in the first week = fluid retention |
| **HemoCue Hb / PCV** | Blood loss (subgaleal), polycythaemia |
| **Bedside clotting (20-minute whole-blood clotting test)** | DIC |

### Step 4: Laboratory & imaging

| Test | When | Expected / abnormal finding | How it changes management | Availability |
|---|---|---|---|---|
| **Cord/arterial blood gas** | At birth/first hour | pH below 7.0, BE 12 or more | Supports intrapartum asphyxia; cooling eligibility in HICs | Few Kampala centres |
| **Electrolytes, urea, creatinine** | Day 1-3 | Hyponatraemia (SIADH), rising creatinine/potassium (AKI) | Fluid restriction; stop potassium; adjust gentamicin | Regional/national referral |
| **Calcium, magnesium** | Seizures | Hypocalcaemia (below 1.75-2.0 mmol/L total) | 10% calcium gluconate 2 mL/kg slow IV | Referral hospitals |
| **FBC, platelets, coagulation** | Bleeding, sepsis | Thrombocytopenia, DIC | FFP, platelets, vitamin K | Referral hospitals |
| **Blood culture, CRP** | Risk factors or signs of sepsis | Positive culture | Antibiotic duration | Referral hospitals |
| **Lumbar puncture** | Fever, bulging fontanelle, seizures without clear HIE, positive culture | Meningitis | 21 days IV antibiotics | Hospital, once stable |
| **LFTs** | Severe HIE | Raised transaminases | Drug dosing, coagulopathy | Referral hospitals |
| **Cranial ultrasound** | Day 1-3 | Oedema, haemorrhage; excludes other lesions | Prognosis | Kampala tertiary |
| **aEEG/EEG** | Seizures, grading | Background suppression, electrographic seizures | Seizure treatment, prognosis | Very limited |

| Test | When | Expected / abnormal finding | How it changes management | Availability |
|---|---|---|---|---|
| **MRI brain** | Day 5-14 | Basal ganglia/thalamus or watershed injury | Best prognostic test | Few private/tertiary centres |

### Step 5: Putting it together

**Model summary line:** "A 14-hour-old term male, birth weight 3.4 kg, born by vacuum after **obstructed labour for 6 hours** with thick meconium and FHR 90/min, needing **4 minutes of bag-mask ventilation** (Apgar 2, 5, 7), now with **moderate HIE (Sarnat 2)**: lethargic, hypotonic, weak suck, constricted pupils, with **two subtle seizures** (lip smacking and eye deviation) treated with phenobarbital. Temperature 37.0 C, glucose 3.8 mmol/L, SpO2 95% in air, no urine yet."

**Problem list:** (1) moderate HIE with seizures; (2) risk of AKI/SIADH (no urine at 14 h); (3) possible meconium aspiration; (4) sepsis to be excluded; (5) feeding (unable to suck); (6) family counselling.

**Worked example: fluids in an oliguric baby.** A 3.0 kg baby with moderate HIE has passed 10 mL of urine in 24 h (0.14 mL/kg/h) and his weight has risen from 3.0 to 3.15 kg. Sodium is 128 mmol/L. Interpretation: **oliguria + weight gain + hyponatraemia = fluid retention (AKI/SIADH)**. Plan: restrict to **insensible losses (about 30-40 mL/kg/day) plus urine output**, no potassium, recheck electrolytes and creatinine, review gentamicin (prolong interval/stop if culture negative), weigh twice daily.

## Differential diagnosis

| Condition | Distinguishing features | Key investigation |
|---|---|---|
| **HIE** | Hypoxic labour, low Apgar, resuscitation, encephalopathy from birth, multi-organ dysfunction | Clinical; gas, imaging |
| **Neonatal sepsis/meningitis** | Maternal fever, ROM over 18 h, fever/hypothermia, bulging fontanelle, onset after a normal period | Culture, CRP, LP |
| **Hypoglycaemia** | Jittery, lethargic, seizures; SGA/IDM; responds to glucose | Glucose |
| **Intracranial/subgaleal haemorrhage** | Instrumental delivery, boggy scalp crossing sutures, falling Hb, shock | Hb, head circumference, cranial US |
| **Maternal drug effect** | Opioid or MgSO4 before delivery; depression improves over hours; no seizures | History; naloxone not routine |
| **Inborn error of metabolism** | Normal at birth, then lethargy, vomiting, seizures after feeds; consanguinity | Glucose, ammonia, lactate, ketones |
| **Neonatal stroke** | Focal seizures in a well-looking term baby, often day 1-3 | Cranial US/MRI |
| **Congenital neuromuscular disease / malformation** | Hypotonia from birth without encephalopathy; dysmorphism, contractures | Genetics, imaging |
| **Hypocalcaemia / hypomagnesaemia** | Jitteriness, seizures, IDM, preterm | Calcium, magnesium |
| **Kernicterus** | Day 3-7, deep jaundice, opisthotonus, high-pitched cry | Bilirubin |

# Investigations

| Test | What you are looking for | Why |
|---|---|---|
| Glucose | Hypo/hyperglycaemia | Both worsen brain injury |
| Electrolytes, creatinine | Hyponatraemia, AKI, hyperkalaemia | Fluids and drug dosing |
| Calcium, magnesium | Hypocalcaemia | Treatable seizure cause |
| FBC, coagulation | DIC, thrombocytopenia, anaemia | Bleeding risk |
| Blood culture, CRP, LP | Sepsis/meningitis | Treatable mimic |
| Cranial US/MRI, EEG | Pattern and severity | Prognosis |

# Management

## 1. Emergency algorithm (first 24 hours)

1. **Resuscitate effectively** (HBB/NRP). Ventilation is the key intervention.
2. **Admit** to the newborn unit/SCBU every baby with encephalopathy, prolonged resuscitation, or abnormal exam.
3. **Thermoregulation**: **36.5-37.5 C**; avoid hyperthermia; no ad-hoc cooling with ice or fans.
4. **Oxygenation and ventilation**: SpO2 **90-95%**; avoid hyperoxia and over-ventilation; bag-mask/CPAP/ventilation for apnoea or respiratory failure.
5. **Circulation**: CRT and HR; normal saline **10 mL/kg** over 10-20 min only if clearly hypovolaemic (e.g. blood loss, subgaleal haemorrhage); dopamine where available for hypotension due to myocardial dysfunction.
6. **Glucose**: keep **2.6 mmol/L or more** (many units target 2.6-8); **10% dextrose 2 mL/kg IV** for hypoglycaemia, then infusion.
7. **Fluids**: **10% dextrose 60 mL/kg/day** day 1; **restrict** (e.g. 40 mL/kg/day or insensible losses + urine output) if oliguric or hyponatraemic.
8. **Seizures**: **phenobarbital** (table below); exclude hypoglycaemia, hypocalcaemia and meningitis.
9. **Antibiotics**: ampicillin + gentamicin if sepsis cannot be excluded; take care with gentamicin in AKI (extend interval, check levels if available).
10. **Feeding**: nil by mouth or small EBM feeds by NGT while unstable; start/advance as the baby improves.
11. **Monitor**: temperature, HR, RR, SpO2, glucose, urine output, weight, Sarnat stage, seizures.
12. **Communicate** with the family; document carefully.

## 2. Seizure management

| Drug | Dose | Route | Frequency | Duration / notes |
|---|---|---|---|---|
| **Phenobarbital** (first line, WHO/UCG) | **Loading 20 mg/kg** | IV slowly over **15-20 min** (IM if no IV) | Once; if seizures persist give **further 10 mg/kg** doses up to **40 mg/kg total** | **Maintenance 5 mg/kg/day** once daily (start 12-24 h after load). Watch for **apnoea and hypotension**; have a bag and mask ready. Stop when seizure-free (often before discharge; senior decision) |
| **Phenytoin** (second line) | **Loading 20 mg/kg** | IV in normal saline over **20-30 min** (max 1 mg/kg/min), with ECG/HR monitoring | Once; maintenance 3-5 mg/kg/day in 2 doses | Not in dextrose (precipitates); arrhythmias |

| Drug | Dose | Route | Frequency | Duration / notes |
|---|---|---|---|---|
| **Levetiracetam** (second line where available) | **40-60 mg/kg** loading | IV/oral | Then 10-30 mg/kg 12-hourly per local protocol | Fewer side effects; less effective than phenobarbital as first line (NEOLEV2 trial) |
| **Midazolam** (refractory, NICU) | 0.05-0.15 mg/kg bolus then infusion | IV | Infusion | Respiratory depression; needs ventilation support |
| **10% calcium gluconate** (if hypocalcaemic) | **2 mL/kg** | **Slow IV** over 10 min with HR monitoring | Then oral/IV maintenance | Extravasation burns; bradycardia if fast |
| **10% dextrose** (if hypoglycaemic) | **2 mL/kg** | IV | Then infusion | Recheck in 30 min |

DRUG DOSING

**Phenobarbital example**: 3.2 kg baby → loading **64 mg** IV over 20 min; if still seizing after 15-30 min, **32 mg** (10 mg/kg), repeat once if needed (total 40 mg/kg = 128 mg); maintenance **16 mg once daily**. Diazepam is **not** recommended as a routine neonatal anticonvulsant (respiratory depression, sodium benzoate displaces bilirubin).

## 3. Fluids and nutrition

| Day | Typical total fluids in HIE | Notes |
|---|---|---|
| Day 1 | **60 mL/kg/day** 10% dextrose (some units 40-60 if oliguric) | Glucose infusion rate ≈ 4 mg/kg/min at 60 mL/kg/day |
| Day 2-3 | Increase by 20-30 mL/kg/day **only if** passing urine and sodium normal | Add sodium after 48 h (e.g. 10% dextrose with 0.18% saline) |
| Oliguria/AKI | **Insensible losses (30-40 mL/kg/day) + urine output** | No potassium; daily weight and electrolytes |

- **Glucose infusion rate (GIR)** = (% dextrose × mL/kg/day) ÷ 144. Example: 10% at 60 mL/kg/day = **4.2 mg/kg/min**.
- **Feeding**: start **EBM by NGT** (10-20 mL/kg/day) once the baby is stable (no seizures for 24 h, abdomen soft, bowel sounds); advance by 20-30 mL/kg/day; move to cup/breast as suck returns. Severe HIE babies may need long-term NGT feeding.

## 4. Temperature and therapeutic hypothermia

- **Therapeutic hypothermia (TH)** = whole-body cooling to **33.5 C for 72 hours**, started **within 6 hours** of birth, followed by slow rewarming (0.5 C per hour), for babies **36 weeks or more** with **moderate or severe HIE**, in a **NICU** with ventilation, monitoring, and blood gases. In high-income settings it reduces death or disability (number needed to treat about 6-9).
- **HELIX trial** (Lancet Global Health 2021; India, Sri Lanka, Bangladesh): TH did **not** reduce death or disability and **significantly increased death** in LMIC tertiary units. Many babies in LMICs have longer, partial insults,

more infection, and different timing, so the benefit from HIC trials does not transfer.

> **UGANDA CONTEXT**
> Therapeutic hypothermia is **not standard care** in Ugandan public hospitals and should be offered only within a research protocol or a fully equipped NICU. On the ward, the priority is the opposite mistake: **avoid hyperthermia** (radiant warmers set too high, heavy wrapping, fever). Keep the baby at **36.5-37.5 C**.

## 5. Other organ support

- **Respiratory**: meconium aspiration → oxygen, CPAP, antibiotics; PPHN (labile SpO2, pre/post-ductal difference) → oxygen, avoid acidosis and hypothermia, refer.
- **Cardiovascular**: hypotension → cautious saline (only if hypovolaemic), inotropes (dopamine 5-10 microgram/kg/min) where available.
- **Renal**: fluid restriction, avoid nephrotoxins, adjust gentamicin; monitor potassium.
- **Liver/blood**: vitamin K, FFP and platelets for bleeding.
- **Gut**: delay or slow feeds; watch for NEC.

## 6. Level-of-care plan in Uganda

| Level | What to do |
|---|---|
| **HC III/HC IV** | HBB resuscitation; keep warm (not hot); breastfeed or EBM by cup if able; check glucose or give EBM/sugar water by NGT; **first dose of phenobarbital 20 mg/kg IM** if convulsing; first dose of ampicillin + gentamicin if risk factors; **refer** with the mother and a referral note |
| **General/regional referral** | SCBU admission, oxygen, IV fluids, glucose monitoring, phenobarbital, antibiotics, NGT feeding, daily Sarnat |
| **National referral/tertiary (Kampala)** | CPAP/ventilation, blood gases, electrolytes, cranial US, EEG/MRI in some centres, multidisciplinary follow-up |

## 7. Prognosis and follow-up

- **Mild HIE**: usually normal outcome; still review development.
- **Moderate HIE**: significant risk of death or disability (roughly a third or more without cooling); outcome depends on recovery.
- **Severe HIE**: high mortality; most survivors have **cerebral palsy** (often spastic quadriplegia or dyskinetic), microcephaly, epilepsy, intellectual disability, cortical visual impairment, hearing loss, feeding problems.
- **Good signs**: normal tone, alertness and **sucking by day 7** (WHO: "an infant who is normally active will usually do well"); seizures controlled early; normal neurological exam at discharge.
- **Bad signs**: severe (Sarnat 3) encephalopathy, seizures that are hard to control, abnormal exam beyond 1 week, **inability to suck at 1 week**, microcephaly at follow-up.
- **Follow-up plan**: weight and head circumference, development at 6 weeks, 3, 6, 9, 12 and 18-24 months; hearing and vision screening; physiotherapy early; nutrition support; epilepsy care; refer to paediatric neurology and cerebral palsy/rehabilitation services (e.g. Mulago, CoRSU, Katalemwa Cheshire Home) and parent support groups.
- **Breaking news**: be honest and gentle; avoid firm predictions on day 1; review with the family daily; involve the father; screen the mother for depression.

# Complications

| Acute | Long-term |
|---|---|
| Seizures, cerebral oedema | **Cerebral palsy** |
| Apnoea, respiratory failure, PPHN, meconium aspiration | Epilepsy |
| AKI, SIADH, hyponatraemia | Intellectual disability, developmental delay |
| Myocardial dysfunction, hypotension | Visual impairment (cortical) |
| Hypoglycaemia, hypocalcaemia | Hearing loss |
| DIC, liver injury | Feeding difficulty, aspiration, malnutrition |
| NEC, feed intolerance | Microcephaly |
| Sepsis | Behavioural problems, family stress |

# Ward-round & exam pearls

## Model presentation

"**Baby Okello**, a **2-day-old** term male, birth weight **3.5 kg**, born at Mengo by emergency CS for **obstructed labour** (referred from a HC IV after 14 hours of labour) with **thick meconium** and **FHR 80/min**. He was **not breathing at birth**, needed **5 minutes of bag-mask ventilation**, HR rose above 100 at 3 minutes; Apgar 2, 4 and 7.

On day 1 he had **moderate HIE (Sarnat 2)**: lethargic, hypotonic, weak suck, constricted pupils, with **three seizures** (cycling movements and eye deviation) controlled with **phenobarbital 20 mg/kg** then a further **10 mg/kg**; now on **5 mg/kg/day**. Temperature 36.8-37.3. He has had no seizures for 20 hours. Glucose 3.5-6.0 mmol/L. On **10% dextrose 60 mL/kg/day**; passed **1.2 mL/kg/h** urine in the last 12 hours. Sodium 132. Blood culture pending; on **ampicillin and gentamicin** (ROM 20 h). Started **EBM 10 mL/kg/day by NGT** today.

Mother **HIV-negative** in labour, RPR negative. **BCG and OPV0 not yet given** (will give before discharge). Weight today **3.4 kg** (–3%).

**Assessment:** moderate HIE, improving, seizures controlled; problems: feeding, fluid balance, sepsis exclusion, family counselling. **Plan:** daily Sarnat, advance feeds as suck returns, stop antibiotics at 48 h if culture negative and well, continue phenobarbital and review, development follow-up at 6 weeks."

## Viva questions (with answers)

1. **What is the difference between birth asphyxia and HIE?** Asphyxia is the event (hypoxia, hypercapnia, acidosis/failure to breathe); HIE is the brain injury that results.
2. **Describe Sarnat stage 2.** Lethargy, hypotonia, weak suck and Moro, constricted pupils, bradycardia, seizures common.
3. **Why avoid hyperthermia?** It increases cerebral metabolic demand and worsens secondary energy failure; even 1 C rise increases the odds of death/disability.
4. **First-line anticonvulsant and dose?** Phenobarbital 20 mg/kg IV over 15-20 min; further 10 mg/kg to 40 mg/kg; maintenance 5 mg/kg/day.
5. **Why restrict fluids in HIE?** Risk of AKI and SIADH → fluid overload, hyponatraemia, cerebral oedema.
6. **Is cooling recommended in Uganda?** Not as routine care; HELIX showed increased mortality in LMICs; use only within protocols in fully equipped NICUs.

7. **What predicts a poor outcome?** Severe encephalopathy, prolonged resuscitation (no HR at 10 min), seizures hard to control, abnormal exam and inability to suck at 1 week, abnormal EEG/MRI.

### OSCE tips

- Expect a station on **Sarnat staging** from a video or description, or on **calculating phenobarbital doses**.
- When counselling parents, use plain words ("the baby's brain was short of oxygen during labour"), avoid blame, explain uncertainty, describe the plan and follow-up.

## Mnemonics & memory aids

**Avoid secondary injury: "Keep it in the middle"**: not too hot, not too cold; not too much glucose, not too little; not too much oxygen, not too little; not too much fluid, not too little.

**Multi-organ injury: "Brain Kills Heart, Lungs, Liver, Gut, Blood"** (brain, kidneys, heart, lungs, liver, gut, blood/DIC).

**Sarnat: "1 = Hyper, 2 = Hypo + fits, 3 = Coma + flat"**.

## High-yield summary

### Numbers & doses to know

| Item | Value |
|---|---|
| Asphyxia supporting criteria | **pH below 7.0**, **BE 12 or more**, Apgar **0-3 beyond 5 min**, encephalopathy, multi-organ failure |
| Latent phase / therapeutic window | About **6 hours** |
| Cooling (where offered) | **33.5 C for 72 h**, start within **6 h**, **36 weeks or more**, moderate/severe HIE |
| HELIX trial (LMIC, 2021) | No benefit; **more deaths** with cooling |
| Temperature target | **36.5-37.5 C**; avoid above 37.5 |
| SpO2 target | **90-95%** |
| Glucose | Treat **below 2.6 mmol/L** with **10% dextrose 2 mL/kg** |
| Day-1 fluids | **10% dextrose 60 mL/kg/day** (restrict if oliguric) |
| Oliguria | **Below 1 mL/kg/h** |
| GIR formula | **(% dextrose × mL/kg/day) ÷ 144** |
| Phenobarbital | **20 mg/kg** load (+10 mg/kg to **40 mg/kg**); maintenance **5 mg/kg/day** |
| Phenytoin | **20 mg/kg** IV over 20-30 min in saline |
| Calcium gluconate 10% | **2 mL/kg** slow IV |
| Saline bolus (if hypovolaemic) | **10 mL/kg** |
| Prognostic checkpoint | Suck and tone at **1 week** |

### Classic exam traps

- **"Apgar of 3 at 1 minute = birth asphyxia."** → Apgar alone does not diagnose asphyxia; it is affected by prematurity, drugs and resuscitation.

- **"Cool all asphyxiated babies with a fan and wet towels."** → Wrong. Uncontrolled cooling is dangerous; **avoid hyperthermia** and offer TH only within NICU protocols.
- **"Give diazepam for neonatal seizures."** → Phenobarbital is first line; diazepam causes apnoea and is not recommended routinely.
- **"Give full maintenance fluids from day 1 and increase daily."** → Restrict if oliguric; AKI and SIADH are common.
- **"No seizures = mild HIE."** → Severe HIE may have few clinical seizures; grade on **consciousness and tone**.
- **"Floppy baby after a long labour: it's HIE; no need for antibiotics."** → Always exclude **sepsis, meningitis and hypoglycaemia**.
- **"Mild HIE needs no follow-up."** → Review development; mild cases can still have subtle problems.

## Questions seniors ask

1. **"What Sarnat stage is he, and why?"** Name each feature (consciousness, tone, suck, pupils, seizures).
2. **"How long was he ventilated and when did the heart rate come up?"** Reflects the depth of the insult.
3. **"Why is he on antibiotics?"** Infection risk (ROM, fever) and sepsis cannot be excluded; stop at 48 h if well and culture negative.
4. **"What is his urine output and fluid plan?"** mL/kg/h and total fluids mL/kg/day.
5. **"What anticonvulsant and how much?"** Phenobarbital in mg/kg and total mg.
6. **"Would you cool this baby?"** Not in our setting outside a NICU protocol (HELIX); keep normothermic.
7. **"What will you tell the parents about the future?"** Honest uncertainty; review suck and tone over the week; follow-up for development, hearing and vision.

SELF-TEST

# Flashcards

26 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | WHO definition of birth asphyxia. | **Failure to initiate and sustain breathing at birth.** Physiologically: impaired gas exchange causing hypoxaemia, hypercapnia and metabolic acidosis. |
| 2 | Difference between neonatal encephalopathy and HIE. | **Neonatal encephalopathy** is the clinical syndrome (altered consciousness, tone, feeding, seizures) in a baby of 35 weeks or more; **HIE** is encephalopathy caused by intrapartum hypoxia-ischaemia. |
| 3 | Criteria supporting an intrapartum hypoxic-ischaemic cause. | Cord/early arterial **pH below 7.0** and/or **base deficit 12 mmol/L or more**; **Apgar 0-3 beyond 5 min** (or below 5 at 5 and 10 min); early moderate/severe encephalopathy; multi-organ dysfunction; sentinel event; compatible imaging. |
| 4 | Sarnat stage 1 (mild) features. | **Hyperalert**, irritable, jittery; normal or increased tone; strong Moro; weak suck; **dilated pupils** (sympathetic); **no seizures**; lasts under 24 h; usually **normal outcome**. |
| 5 | Sarnat stage 2 (moderate) features. | **Lethargic**, **mild hypotonia**, weak Moro and suck, strong distal flexion, **constricted pupils**, bradycardia (parasympathetic), **seizures common**; lasts 2-14 days; variable outcome. |
| 6 | Sarnat stage 3 (severe) features. | **Stupor/coma**, **flaccid**, absent reflexes (Moro, suck), decerebrate posturing, variable/poorly reactive pupils, apnoea; seizures may be suppressed; **high mortality, survivors usually disabled**. |
| 7 | What happens in the latent phase of HIE and why does it matter? | After resuscitation, brain energy partly recovers for about **6 hours** before **secondary energy failure** (6-72 h). This is the **therapeutic window**: cooling must start within 6 h; supportive care aims to prevent secondary injury. |
| 8 | Name six factors that worsen secondary brain injury. | **Hyperthermia**, **hypoglycaemia** (and hyperglycaemia), **seizures**, **hypotension**, hypoxaemia/**hyperoxia**, **hypocapnia** (over-ventilation), infection, fluid overload. |
| 9 | Target temperature for a baby with HIE in a Ugandan ward. | **36.5-37.5 C**. Avoid **hyperthermia** (radiant warmer too high, overwrapping). Do not actively cool outside a NICU protocol. |
| 10 | Therapeutic hypothermia protocol (where offered). | Whole-body cooling to **33.5 C for 72 h**, started **within 6 h** of birth, for babies **36 weeks or more** with **moderate/severe HIE**, then rewarm at 0.5 C/h; needs NICU support. |
| 11 | What did the HELIX trial show? | In tertiary units in **India, Sri Lanka and Bangladesh** (Lancet Global Health 2021), cooling **did not reduce death or disability** and **significantly increased death**. So cooling is not standard care in LMICs like Uganda. |
| 12 | Phenobarbital regimen for neonatal seizures. | **Loading 20 mg/kg IV** over 15-20 min; if seizures persist, further **10 mg/kg** doses to **40 mg/kg total**; maintenance **5 mg/kg/day once daily**. Watch for apnoea. |
| 13 | Second-line anticonvulsants in neonates. | **Phenytoin 20 mg/kg IV** over 20-30 min in saline (cardiac monitoring), or **levetiracetam 40-60 mg/kg**, or midazolam (NICU). Diazepam is not routine (apnoea, displaces bilirubin). |
| 14 | Day-1 fluids in HIE. | **10% dextrose 60 mL/kg/day**, restricted (e.g. 40 mL/kg/day or insensible losses + urine output) if **oliguric or hyponatraemic**. |

| # | Question | Answer |
|---|---|---|
| 15 | Define oliguria in a neonate. | Urine output **below 1 mL/kg/h** (after the first 24 h). Weigh nappies: 1 g = 1 mL. |
| 16 | Glucose infusion rate (GIR) formula. | **GIR (mg/kg/min) = (% dextrose × mL/kg/day) ÷ 144.** 10% at 60 mL/kg/day = 4.2 mg/kg/min. |
| 17 | Why does asphyxia injure the kidneys and gut? | The **diving reflex** redistributes blood to the **brain, heart and adrenals**, starving the kidneys, gut, liver, skin and lungs → AKI, NEC/feed intolerance, liver injury. |
| 18 | Treatment of hypocalcaemic seizures in a neonate. | **10% calcium gluconate 2 mL/kg slow IV** over about 10 min with heart rate monitoring (risk of bradycardia and extravasation burns), then maintenance. |
| 19 | WHO clinical prognostic rule for HIE. | A baby who **within a week is still floppy or spastic, unresponsive and cannot suck** has severe brain injury and will do poorly; one who is normally active usually does well. |
| 20 | Main long-term complications of HIE. | **Cerebral palsy** (spastic quadriplegia, dyskinetic), **epilepsy**, intellectual disability, cortical visual impairment, hearing loss, microcephaly, feeding difficulty. |
| 21 | Basal ganglia/thalamus vs watershed injury pattern. | **Acute profound** insult (cord prolapse, rupture, abruption) → basal ganglia/thalami → dyskinetic/spastic CP. **Prolonged partial** insult (long obstructed labour) → watershed cortex/white matter → cognitive problems, spastic CP. |
| 22 | Pupils in mild vs moderate HIE. | Mild: **dilated** (sympathetic overactivity). Moderate: **constricted** (parasympathetic), with bradycardia and secretions. |
| 23 | What should a HC III/IV do for an asphyxiated baby before referral? | HBB resuscitation; keep warm (not hot); glucose check or EBM/sugar water by cup/NGT; **phenobarbital 20 mg/kg** (IV/IM) if convulsing; first dose of **ampicillin + gentamicin** if risk factors; refer with the mother and a note. |
| 24 | Important mimics of HIE to exclude. | **Sepsis/meningitis**, **hypoglycaemia**, intracranial or **subgaleal haemorrhage**, maternal drugs (opioids, MgSO4), hypocalcaemia, inborn errors of metabolism, neonatal stroke, neuromuscular disease. |
| 25 | When do HIE seizures usually start? | Usually **within the first 12-24 hours** of life; often **subtle** (lip smacking, eye deviation, cycling, apnoea). |
| 26 | Thompson score. | A 0-22 bedside HIE score (tone, consciousness, fits, posture, Moro, grasp, suck, respiration, fontanelle) used in LMIC research and some Ugandan units; higher = worse; about 15 or more predicts poor outcome. |

SELF-TEST

# Single best answer questions

15 questions. The answers and explanations follow at the end.

**1.** A term baby born after a 3-hour second stage needed 4 minutes of bag-mask ventilation. At 12 hours he is lethargic, hypotonic, with a weak suck, constricted pupils and an episode of lip smacking with eye deviation. Which Sarnat stage is he?

A Stage 1 (mild)
B Normal post-resuscitation state
C Stage 2 (moderate)
D Stage 3 (severe)
E Cannot be staged without an EEG

**2.** A 3.0 kg term baby with moderate HIE has a generalised clonic seizure at 8 hours. Glucose is 4.1 mmol/L. IV access is in place. What is the MOST appropriate first drug?

A Phenobarbital 60 mg IV over 15-20 minutes
B Diazepam 3 mg IV repeated every 5 minutes
C Phenytoin 60 mg IV push over 1 minute
D 10% dextrose 6 mL IV
E Paracetamol 45 mg rectally

**3.** A baby with moderate HIE is under a radiant warmer on day 1. The nurse records an axillary temperature of 38.2 C. There are no other signs of infection. What is the MOST appropriate immediate action?

A Increase the radiant warmer to prevent shivering
B Give paracetamol 15 mg/kg and continue as before
C Apply ice packs and a fan to bring the temperature to 33.5 C
D Reduce/turn off the warmer, remove extra wraps, recheck in 15-30 min, and consider sepsis
E Ignore it; hyperthermia is protective after asphyxia

**4.** A 3.2 kg baby with moderate HIE on 10% dextrose 60 mL/kg/day has passed 15 mL of urine in 24 hours. Weight is now 3.35 kg, sodium 127 mmol/L, and he has mild periorbital oedema. What is the MOST appropriate fluid plan?

A Increase fluids to 90 mL/kg/day as per day-2 routine
B Give furosemide and increase fluids to 120 mL/kg/day
C Give normal saline 20 mL/kg bolus to improve kidney perfusion
D Add potassium chloride to the infusion
E Restrict fluids to insensible losses plus urine output, avoid potassium, recheck electrolytes and creatinine

**5.** Which statement about therapeutic hypothermia for HIE is MOST accurate for a Ugandan regional referral hospital?

A It should be started at any time in the first week
B It is not standard care; in LMIC units the HELIX trial found no benefit and more deaths, so the priority is to avoid hyperthermia
C It is first-line care for all asphyxiated babies, including preterm babies below 34 weeks
D Passive cooling with fans is a safe alternative when no device is available
E It cures HIE if started within 24 hours

**6.** A term baby is floppy and lethargic at 30 hours. Labour was normal and short; Apgar 9 and 10. The mother had a fever of 38.6 C in labour and membranes ruptured 22 hours before birth. The fontanelle is full. What is the MOST likely diagnosis?

A Mild HIE
B Severe HIE
C Neonatal sepsis with possible meningitis
D Maternal opioid effect
E Benign neonatal sleep myoclonus

**7.** A baby born by vacuum extraction after a prolonged second stage becomes pale and tachycardic at 4 hours. There is a boggy, fluctuant scalp swelling crossing the suture lines, extending to the neck, and the head circumference has increased by 2 cm. What is the MOST appropriate next step?

A Aspirate the swelling to reduce pressure
B Reassure; it is caput succedaneum and will resolve in 48 hours
C Apply a tight head bandage and continue routine care
D Start phenobarbital for presumed HIE
E Treat as subgaleal haemorrhage: IV access, saline 10 mL/kg and urgent blood, vitamin K, check Hb and clotting, monitor head circumference

**8.** A 3.5 kg baby with HIE continues to have seizures 30 minutes after a 70 mg phenobarbital loading dose. Glucose is 3.8 mmol/L. What is the MOST appropriate next step?

A Give another phenobarbital dose of 35 mg (10 mg/kg) IV slowly, with a bag and mask ready
B Give a second full 70 mg dose immediately as an IV push
C Give diazepam 5 mg IV
D Wait 24 hours for the phenobarbital to work
E Give 10% dextrose 2 mL/kg

**9.** Which finding at 7 days of life BEST predicts a poor neurological outcome in a term baby with HIE, when MRI and EEG are not available?

A A single seizure on day 1 that was controlled with phenobarbital
B Physiological jaundice on day 4
C Still unresponsive, floppy or spastic, and unable to suck
D Weight loss of 7% from birth weight
E Apgar 4 at 1 minute

**10.** After asphyxia, blood is redirected to protect certain organs. Which organ is MOST protected by this diving reflex?

A Kidneys
B Gut
C Skin
D Adrenal glands
E Liver

**11.** A term baby at a HC IV needed 3 minutes of bag-mask ventilation, now breathes at 50/min but is lethargic with a weak suck. The nearest SCBU is 2 hours away. Which plan is MOST appropriate before referral?

A Discharge home with the mother; return if seizures

B Give expressed colostrum by cup/NGT or check glucose, keep normothermic (skin-to-skin), and refer with a referral note and the mother

C Wrap in three blankets under a heater because asphyxiated babies get cold

D Give phenobarbital 20 mg/kg as prophylaxis even though there are no seizures

E Give sodium bicarbonate to correct presumed acidosis

**12.** Which pattern of brain injury is MOST typical after an acute, near-total insult such as a cord prolapse with 25 minutes of profound bradycardia?

A Parasagittal watershed injury only

B Periventricular leukomalacia

C Isolated germinal matrix haemorrhage

D Basal ganglia and thalamic injury

E Cerebellar hypoplasia

**13.** A baby with moderate HIE on day 2 has a seizure. Glucose 3.9 mmol/L, total calcium 1.5 mmol/L (low). He has already received 40 mg/kg phenobarbital. What is the MOST appropriate next step?

A 10% calcium gluconate 2 mL/kg slow IV over 10 minutes with heart-rate monitoring

B Another 20 mg/kg phenobarbital

C 10% calcium gluconate 2 mL/kg as a rapid IV push

D Oral calcium only

E Magnesium sulphate 50% 1 mL/kg IV push

**14.** When is it MOST appropriate to start enteral feeds in a baby with moderate HIE?

A Immediately at birth, full volume by bottle

B Only after discharge

C After 2 weeks of IV fluids

D When stable: no seizures for about 24 hours, abdomen soft with bowel sounds; start small EBM feeds by NGT and advance as tolerated

E Only when the baby can suck at the breast

**15.** Which statement about the Apgar score is CORRECT?

A A 1-minute Apgar of 3 confirms birth asphyxia

B Resuscitation should start after the 1-minute Apgar is recorded

C The Apgar score describes the baby's condition but alone does not diagnose asphyxia; an Apgar of 0-3 beyond 5 minutes supports a hypoxic cause

D A 5-minute Apgar of 7 rules out HIE

E Apgar scores are not affected by prematurity or maternal drugs

## Answers & explanations

> **1. C** · Stage 2 (moderate)
>
> C is right: lethargy, hypotonia, weak suck, constricted pupils (parasympathetic) and seizures (the lip smacking and eye deviation are subtle seizures) define Sarnat stage 2. A is wrong: stage 1 babies are hyperalert with normal or increased tone, dilated pupils and no seizures. B is wrong: encephalopathy with seizures is not normal. D is wrong: stage 3 babies are stuporous/comatose and flaccid with absent reflexes. E is wrong: Sarnat staging is clinical; EEG adds detail but is not required.

**2. A** · Phenobarbital 60 mg IV over 15-20 minutes

A is right: phenobarbital is first line (WHO/UCG); 20 mg/kg × 3.0 kg = 60 mg IV slowly over 15-20 min, with extra 10 mg/kg doses to 40 mg/kg if needed. B is wrong: diazepam is not recommended routinely in neonates (apnoea) and 1 mg/kg repeated is an overdose. C is wrong: phenytoin is second line and must be given slowly (max 1 mg/kg/min) with cardiac monitoring. D is wrong: glucose is normal, so dextrose will not stop the seizure. E is wrong: there is no fever, and paracetamol is not an anticonvulsant.

**3. D** · Reduce/turn off the warmer, remove extra wraps, recheck in 15-30 min, and consider sepsis

D is right: hyperthermia worsens secondary brain injury after asphyxia; the commonest cause on the ward is environmental (warmer/wraps). Correct it and recheck, while keeping sepsis in mind. A is wrong: it makes hyperthermia worse. B is wrong: antipyretics are not used in neonates (WHO: control the environment) and do not address the cause. C is wrong: uncontrolled cooling with ice and fans outside a NICU protocol is dangerous, and HELIX showed increased mortality with cooling in LMICs. E is wrong: hyperthermia is harmful, not protective.

**4. E** · Restrict fluids to insensible losses plus urine output, avoid potassium, recheck electrolytes and creatinine

E is right: urine output is about 0.2 mL/kg/h (oliguria), with weight gain, oedema and hyponatraemia: AKI and/or SIADH with fluid retention. Restrict to insensible losses (about 30-40 mL/kg/day) plus urine output, avoid potassium, monitor creatinine and sodium. A is wrong: routine increases cause fluid overload and cerebral oedema. B is wrong: increasing fluids is the opposite of what is needed. C is wrong: he is overloaded, not hypovolaemic. D is wrong: potassium is dangerous in AKI (hyperkalaemia).

**5. B** · It is not standard care; in LMIC units the HELIX trial found no benefit and more deaths, so the priority is to avoid hyperthermia

B is right: HELIX (India, Sri Lanka, Bangladesh, 2021) found that cooling did not reduce death or disability and increased mortality; cooling is offered only within NICU protocols, and the ward priority is normothermia (36.5-37.5 C). A is wrong: the window is within 6 hours. C is wrong: cooling trials were in babies of 36 weeks or more with moderate/severe HIE. D is wrong: uncontrolled cooling causes overcooling and harm. E is wrong: even in HICs cooling reduces, not abolishes, death/disability (NNT about 6-9), and it must start within 6 h.

**6. C** · Neonatal sepsis with possible meningitis

C is right: a normal labour with good Apgar scores makes HIE unlikely; maternal fever, prolonged ROM and a full fontanelle point to early-onset sepsis/meningitis. Take cultures, do an LP when stable, and start ampicillin + gentamicin (meningitis doses). A and B are wrong: HIE requires a hypoxic insult and is present from birth; this baby was vigorous. D is wrong: opioid depression is present at birth and wears off over hours, not starting at 30 h. E is wrong: benign sleep myoclonus occurs in a well, alert baby only during sleep.

**7. E** · Treat as subgaleal haemorrhage: IV access, saline 10 mL/kg and urgent blood, vitamin K, check Hb and clotting, monitor head circumference

E is right: a fluctuant swelling crossing sutures after vacuum, with pallor, tachycardia and rising head circumference, is a subgaleal haemorrhage, which can hold most of a baby's blood volume. Resuscitate with volume and blood, give vitamin K and correct coagulopathy. A is wrong: aspiration introduces infection and does not stop bleeding. B is wrong: caput is a soft oedematous swelling present at birth without shock. C is wrong: bandaging does not control the bleed and delays resuscitation. D is wrong: the problem is haemorrhagic shock, not HIE seizures.

**8. A** · Give another phenobarbital dose of 35 mg (10 mg/kg) IV slowly, with a bag and mask ready

A is right: if seizures persist after the 20 mg/kg load, WHO advises further 10 mg/kg doses up to 40 mg/kg total (here 35 mg slowly), watching for apnoea. B is wrong: a second 20 mg/kg as a push risks apnoea and hypotension; doses should be given slowly in 10 mg/kg steps. C is wrong: 5 mg diazepam (about 1.4 mg/kg) is an overdose and diazepam is not routine in neonates. D is wrong: ongoing seizures worsen brain injury and must be treated. E is wrong: glucose is normal.

**9. C** · Still unresponsive, floppy or spastic, and unable to suck

C is right: the WHO Pocket Book states that an infant who within a week is still floppy or spastic, unresponsive and cannot suck has severe brain injury and will do poorly. A is wrong: a single early, controlled seizure is less predictive. B is wrong: physiological jaundice is unrelated. D is wrong: 7% weight loss is within normal limits. E is wrong: the 1-minute Apgar has poor prognostic value; prolonged low Apgar (e.g. 0-3 at 10 min) is more meaningful.

**10. D** · Adrenal glands

D is right: during hypoxia, blood is redistributed to the brain, heart and adrenals. A, B, C and E are wrong: the kidneys (AKI), gut (NEC, feed intolerance), skin and liver (raised enzymes, coagulopathy, hypoglycaemia) are sacrificed, which explains multi-organ dysfunction after asphyxia.

**11. B** · Give expressed colostrum by cup/NGT or check glucose, keep normothermic (skin-to-skin), and refer with a referral note and the mother

B is right: an encephalopathic baby needs SCBU care; before referral, prevent hypoglycaemia (EBM or check glucose), keep 36.5-37.5 C (skin-to-skin), give antibiotics if risk factors, and refer with documentation. A is wrong: a lethargic baby with a weak suck after resuscitation must not go home. C is wrong: overheating worsens brain injury. D is wrong: prophylactic phenobarbital is not recommended without seizures. E is wrong: sodium bicarbonate is listed by UCG as a harmful practice in newborn resuscitation.

**12. D** · Basal ganglia and thalamic injury

D is right: an acute profound insult damages the most metabolically active deep grey matter (basal ganglia and thalami), leading to dyskinetic or spastic quadriplegic CP. A is wrong: watershed injury follows prolonged partial insults. B and C are wrong: PVL and germinal matrix haemorrhage are injuries of preterm babies. E is wrong: cerebellar hypoplasia is developmental, not an acute hypoxic pattern.

**13. A** · 10% calcium gluconate 2 mL/kg slow IV over 10 minutes with heart-rate monitoring

A is right: hypocalcaemia is a treatable cause of seizures after asphyxia; WHO advises 10% calcium gluconate 2 mL/kg as a slow IV infusion, monitoring heart rate, then supplementation. B is wrong: he has reached the 40 mg/kg maximum and the cause is metabolic. C is wrong: a rapid push causes bradycardia/arrhythmia and extravasation burns. D is wrong: oral calcium is too slow for an acute seizure. E is wrong: magnesium may be needed if hypomagnesaemic, but 50% undiluted 1 mL/kg as a push is dangerous and not first line here.

**14. D** · When stable: no seizures for about 24 hours, abdomen soft with bowel sounds; start small EBM feeds by NGT and advance as tolerated

D is right: once stable, start small volumes of expressed breast milk by NGT (e.g. 10-20 mL/kg/day) and advance by 20-30 mL/kg/day, moving to cup/breast as the suck returns. A is wrong: an encephalopathic baby cannot coordinate feeds safely, and the gut may be ischaemic. B and C are wrong: prolonged IV fluids increase infection risk and delay recovery. E is wrong: waiting for a good suck may take days to weeks; NGT feeding allows nutrition meanwhile.

**15. C** · The Apgar score describes the baby's condition but alone does not diagnose asphyxia; an Apgar of 0-3 beyond 5 minutes supports a hypoxic cause

C is right: Apgar is a description that is affected by many factors; prolonged low scores support (but do not prove) an intrapartum hypoxic cause. A is wrong: a single low 1-minute score is common and often recovers. B is wrong: resuscitation must start within the Golden Minute, never waiting for the Apgar. D is wrong: some babies with HIE have reasonable 5-minute scores, especially after effective resuscitation. E is wrong: prematurity, maternal sedation and congenital anomalies lower Apgar scores.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: Obstructed labour, a non-breathing baby and seizures

A 19-year-old primigravida is referred from a HC III in Wakiso to Mengo after 16 hours of labour with no progress. She has an emergency CS for obstructed labour. The baby boy is born with thick meconium, floppy, not breathing, HR 60/min. He is ventilated with bag and mask; HR rises above 100 at 4 minutes and he gasps at 6 minutes. Apgar 1, 3, 6. Birth weight 3.3 kg.

**1. After resuscitation, what are your immediate priorities on arrival in the newborn unit?**

> Airway and breathing (SpO2 90-95%, avoid hyperoxia; CPAP/bag-mask if apnoea), **temperature 36.5-37.5 C** (avoid overheating), **glucose** now, IV access with **10% dextrose 60 mL/kg/day** (≈ 8.3 mL/h), check perfusion; observe for seizures; vitamin K; inform the senior and the mother.

**2. At 6 hours he is lethargic, hypotonic, weak suck, pupils constricted, HR 105. Stage him and list what you will monitor.**

> **Sarnat stage 2 (moderate HIE).** Monitor 1-2-hourly: temperature, HR, RR/apnoea, SpO2, seizures; glucose 3-6-hourly; urine output (weigh nappies), daily weight, electrolytes/creatinine where available; repeat Sarnat at least daily.

**3. At 10 hours he has cycling movements of the legs and eye deviation for 2 minutes. What will you do (with doses)?**

> Airway, oxygen, **check glucose** (and calcium). Give **phenobarbital 20 mg/kg = 66 mg IV over 15-20 min**; if seizures continue, **10 mg/kg = 33 mg** up to a total of **40 mg/kg (132 mg)**; maintenance **5 mg/kg/day = 16.5 mg once daily** from 12-24 h. Bag and mask ready for apnoea.

**4. What else could be causing or contributing to his encephalopathy, and how will you check?**

> **Sepsis/meningitis** (long labour, multiple vaginal examinations at HC III: blood culture, CRP, LP when stable; start ampicillin + gentamicin), **hypoglycaemia** (glucose), **hypocalcaemia** (calcium), **intracranial or subgaleal haemorrhage** (head exam, Hb), meconium aspiration with hypoxaemia (CXR, SpO2).

**5. On day 2 urine output is 0.4 mL/kg/h and sodium 128 mmol/L. How do you adjust his care?**

> **AKI/SIADH**: restrict fluids to insensible losses + urine output (about 30-40 mL/kg/day + measured output), no potassium, recheck electrolytes and creatinine, **extend gentamicin interval or stop it** if cultures negative at 48 h, weigh twice daily.

**6. On day 7 he is alert, tone improving and sucking from a cup. How do you counsel the family about prognosis and follow-up?**

> Improvement in alertness, tone and suck by day 7 is **encouraging**, but moderate HIE carries a risk of CP, epilepsy and developmental delay. Plan: continue breastfeeding/cup feeds, review at 6 weeks and then 3, 6, 9, 12 and 18 months for development, head growth, hearing and vision; early physiotherapy if tone abnormal; decide on stopping phenobarbital (senior); immunise (BCG/OPV0). Be honest and supportive; give danger signs.

**7. What could have prevented this?**

> Partograph use and **early referral** when the alert/action lines were crossed, timely CS, fetal heart monitoring, avoiding long delays in transport, and skilled resuscitation at birth. Discuss in the perinatal death/near-miss (MPDSR) review without blame.

> **Take-away:** In moderate HIE, keep the baby normothermic, glycaemic and well ventilated, treat seizures with phenobarbital in mg/kg, restrict fluids when urine output falls, exclude sepsis, and judge prognosis by recovery of suck and tone over the first week.

## Case 2: A floppy baby after vacuum delivery

At Kawempe National Referral Hospital, a 4.0 kg baby girl was born by vacuum extraction after a 2.5-hour second stage. She cried at 2 minutes after stimulation (Apgar 5 and 8). At 3 hours the midwife calls you because she is pale and 'not active'. HR 182/min, RR 64/min, CRT 4 seconds, temperature 36.8 C. Glucose 3.2 mmol/L.

**1. What examination finding are you specifically looking for, and why?**

> A **boggy, fluctuant scalp swelling crossing suture lines**, spreading to the forehead/neck, with **increasing head circumference** = **subgaleal haemorrhage** (vacuum), which can cause hypovolaemic shock. Compare with caput (soft oedema, crosses sutures, present at birth, no shock) and cephalhaematoma (bounded by sutures, no shock).

**2. You find a fluctuant swelling over the whole scalp. What is your immediate management?**

> **Call for help**; IV/IO access; **normal saline 10 mL/kg (40 mL)** over 5-10 min, repeat as needed; urgent **cross-match**, give **O-negative/group-specific blood** 10-20 mL/kg; **vitamin K**; check Hb, platelets, clotting (FFP if DIC); monitor HR, CRT, head circumference hourly; keep warm; nil by mouth initially.

**3. How do you distinguish encephalopathy due to shock from HIE?**

> HIE: hypoxic insult before/at birth, needs resuscitation, encephalopathy from birth. This baby was relatively vigorous (Apgar 8 at 5 min) and deteriorated later with **signs of shock** (tachycardia, prolonged CRT, pallor): **hypovolaemia** from bleeding is the main cause, though shock can cause secondary hypoxic-ischaemic injury.

**4. What complications will you monitor for over the next days?**

> **Hypovolaemic shock**, DIC, **jaundice** (breakdown of the haematoma: check bilirubin), seizures, AKI, infection. Monitor Hb, bilirubin, urine output, neurology.

**5. How do you counsel the parents?**

> Explain that bleeding happened under the scalp after the vacuum delivery, what you are doing (fluids, blood), that the swelling resolves over weeks, the risk of jaundice and the need for observation, and follow-up for development. Document and report through clinical audit.

> **Take-away:** Not every floppy baby after a difficult delivery has HIE: a pale, tachycardic baby after vacuum delivery may be bleeding into a subgaleal haemorrhage and needs volume and blood immediately.

## Case 3: Day 5: is this still asphyxia?

A term 3.1 kg baby boy admitted with mild HIE (hyperalert, jittery, no seizures) after a prolonged labour is ready for discharge on day 3. On day 5 the mother returns to Mengo because he has stopped feeding and is 'too sleepy'. Temperature 38.3 C, HR 176, RR 66, mild chest indrawing, fontanelle full. Glucose 2.4 mmol/L.

**1. What is the most likely diagnosis and why is it not worsening HIE?**

> **Late-onset neonatal sepsis, possibly meningitis.** HIE is maximal in the first 72 h and then improves; a baby who was improving and then deteriorates with **fever, tachycardia, tachypnoea and a full fontanelle** has a new problem.

**2. What are your first actions?**

> ABC; oxygen if SpO2 below 90%; **10% dextrose 2 mL/kg = 6 mL IV** and recheck in 30 min; maintenance glucose; blood culture; start antibiotics **without delay**; LP when stable (within 2 h of antibiotics ideally).

**3. Which antibiotics and doses (day 5, 3.1 kg)?**

> Meningitis doses: **ampicillin 100 mg/kg = 310 mg IV 12-hourly** (first week; 8-hourly after day 7) + **gentamicin 5 mg/kg = 15.5 mg IV once daily**; or per UCG/WHO, **ceftriaxone 50 mg/kg 12-hourly or 100 mg/kg once daily** with gentamicin if meningitis likely (avoid ceftriaxone if jaundiced or with IV calcium). Duration **21 days** if meningitis confirmed.

**4. What supportive care does he need?**

> Temperature control (undress, no paracetamol in neonates per UCG), fluids 10% dextrose with careful volume, NGT EBM when stable, monitor glucose, seizures (phenobarbital if needed), SpO2, urine output.

**5. What is the lesson for follow-up of babies with HIE?**

> Discharge counselling must include **danger signs** (not feeding, fever, fast breathing, convulsions, sleepiness), and any deterioration after initial improvement should be treated as a **new illness**, most often sepsis.

> **Take-away:** HIE improves after the first 72 hours; a new deterioration with fever or a full fontanelle is sepsis or meningitis until proven otherwise.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.